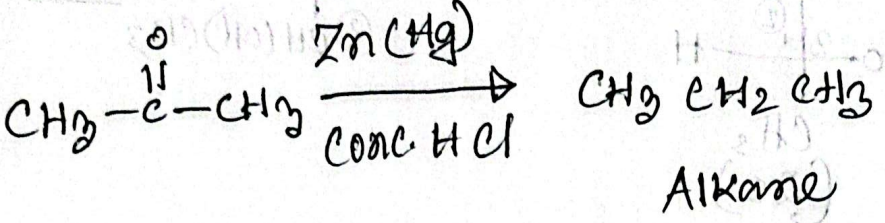
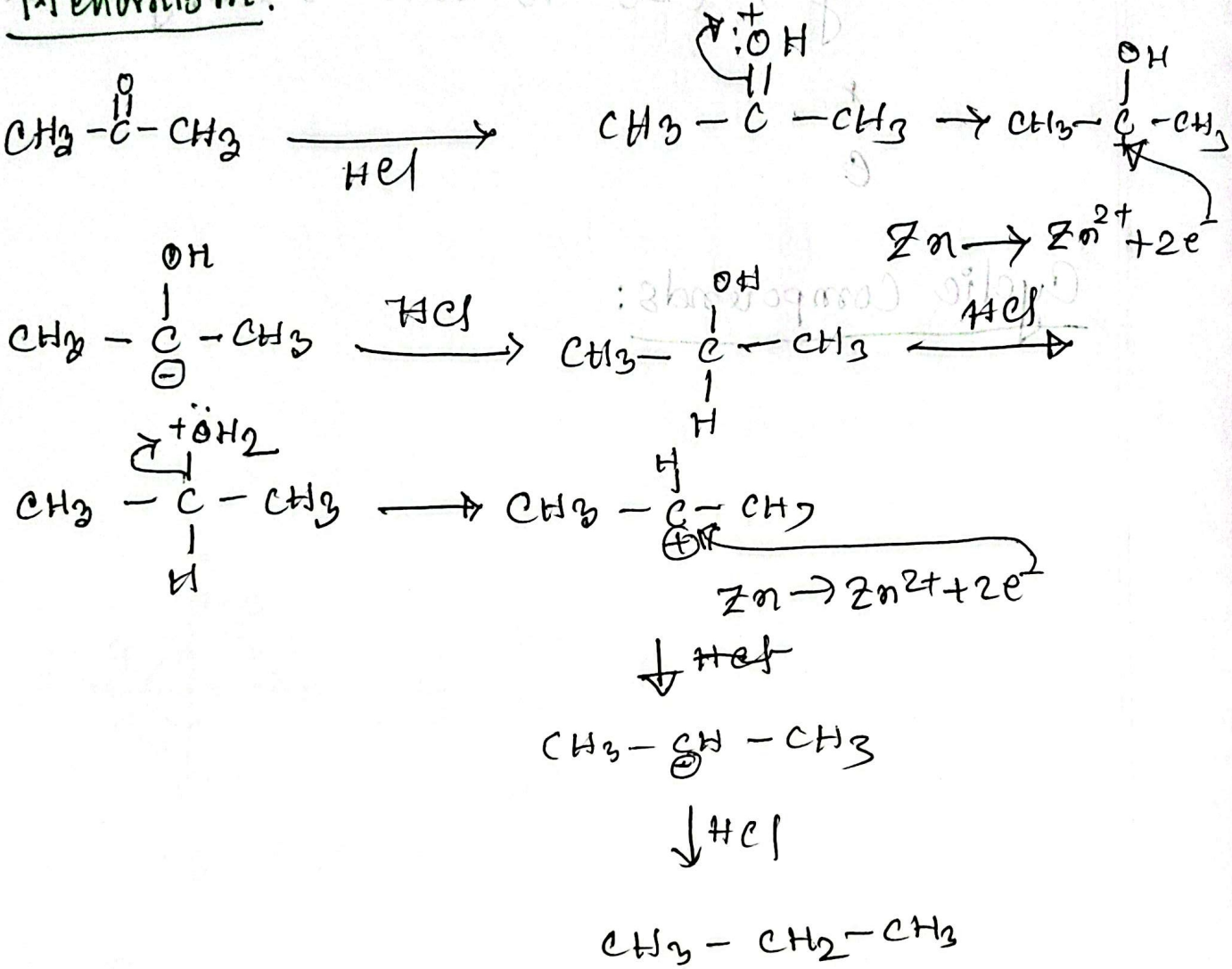
Name Reactions SUZON-50

Clemmensen Reduction

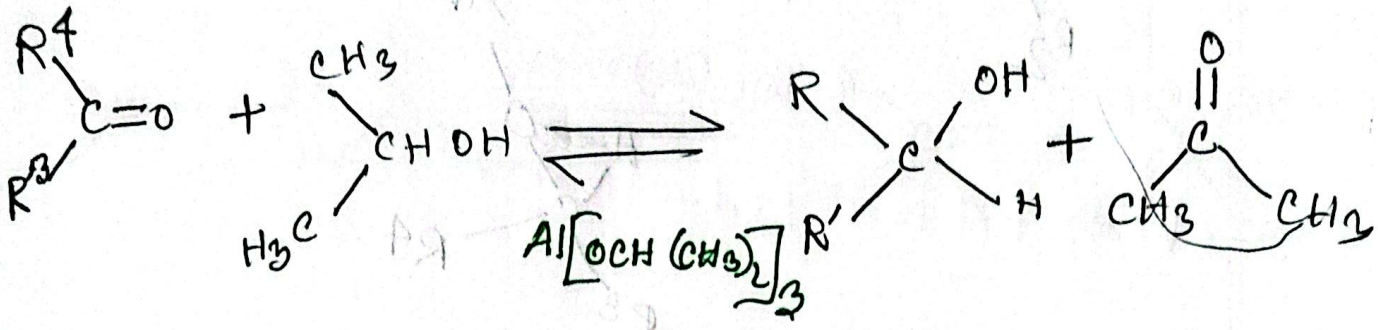


Mechanism:

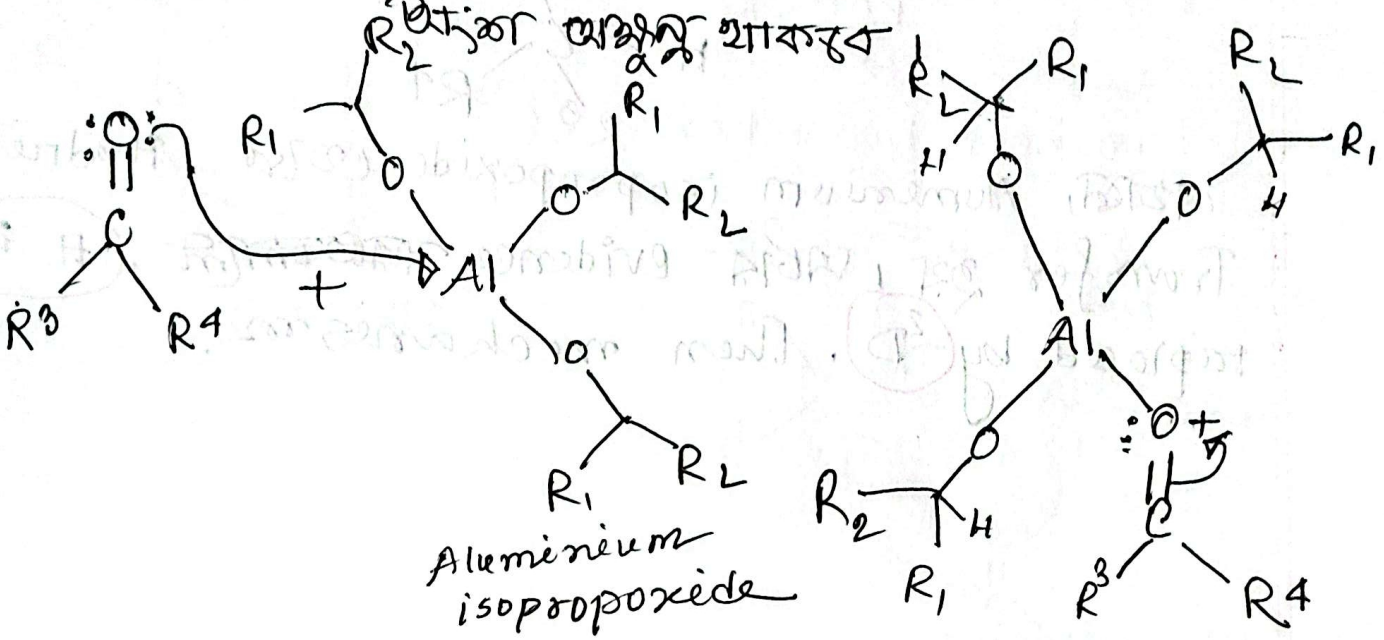


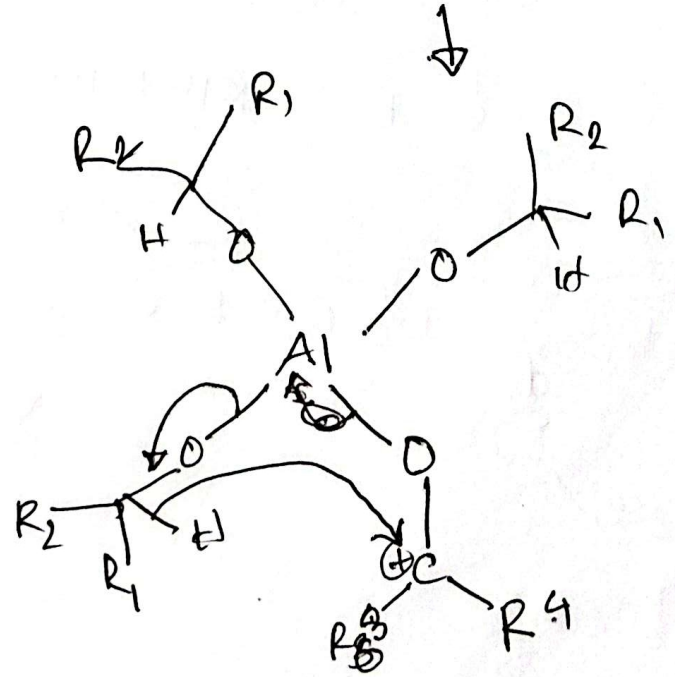
Applicat

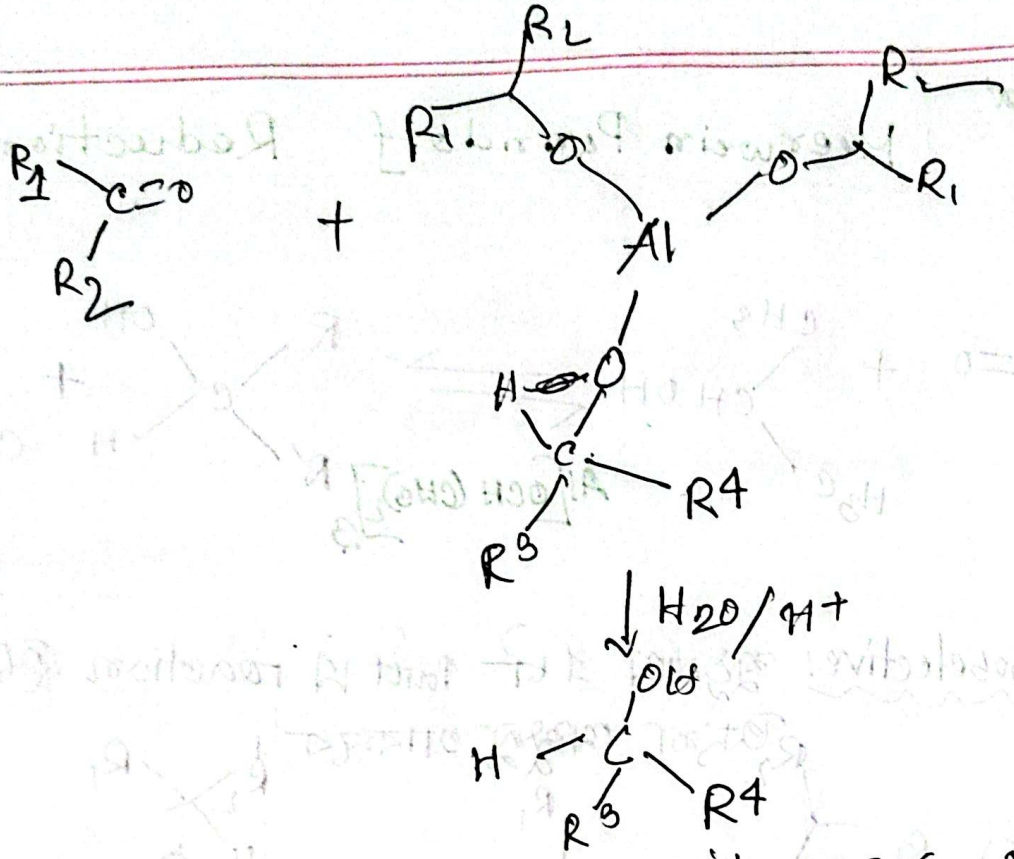
Meerwein Ponndorf Reduction



Chemoselective! শুধুমাত্র 1 টি Part এ reaction করবে। অপর R2 অংশ অক্ষুন্ন থাকবে

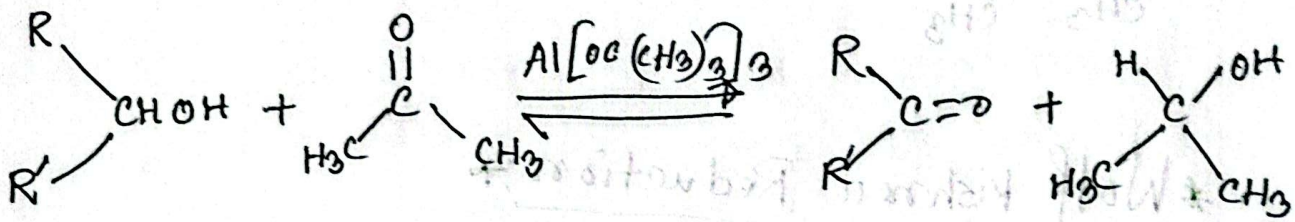


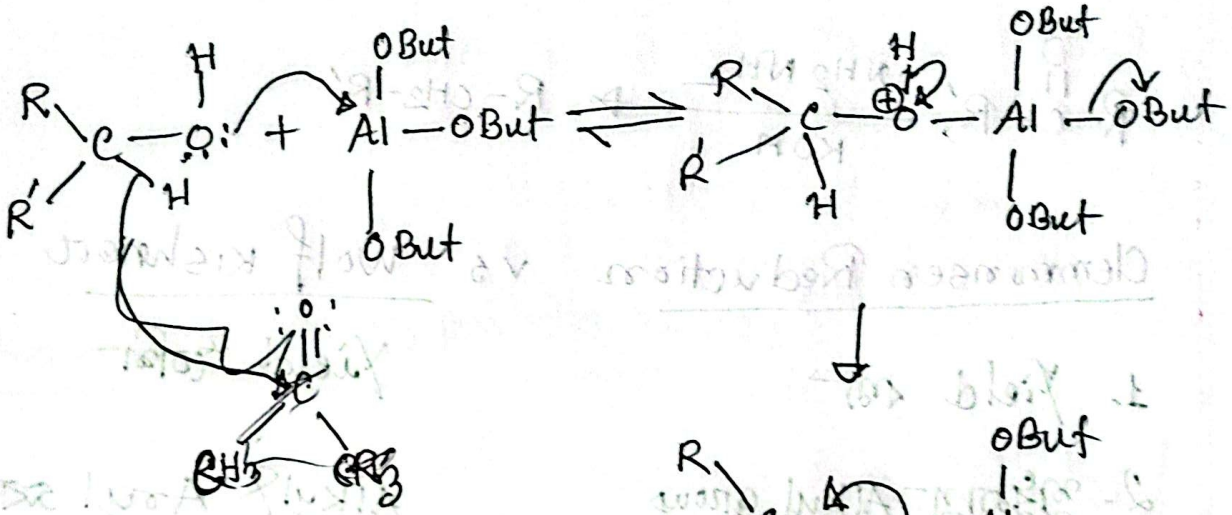


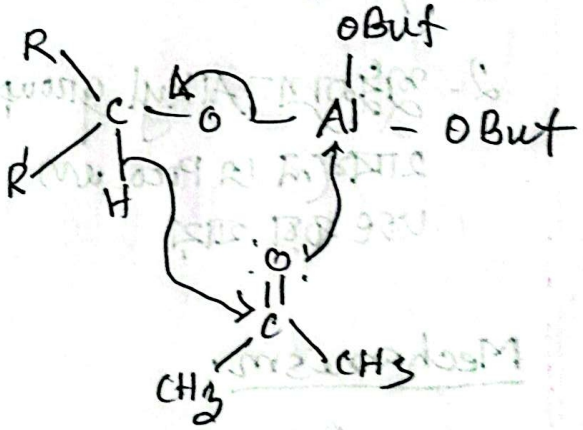


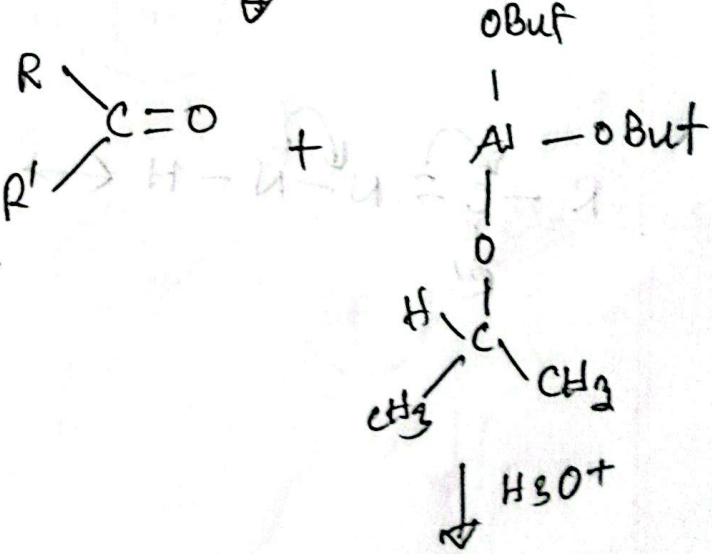
এখানে, Aluminium isopropoxide থেকে Hydride Transfer হয়। এটার evidence [illegible], H is replaced by D. Then mechanism.

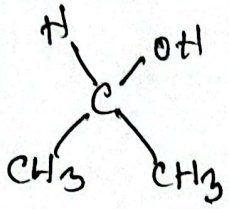
Oppenauer Oxidation:











* Wolf kishner Reduction *

$$R-\overset{O}{\overset{\|}{C}}-R' \xrightarrow[KOH]{NH_2NH_2} R-CH_2-R'$$

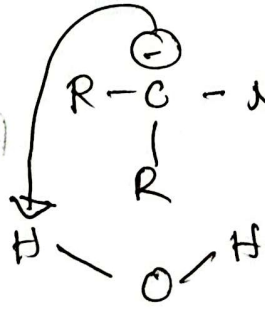
Clemmensen Reduction vs Wolf kishner

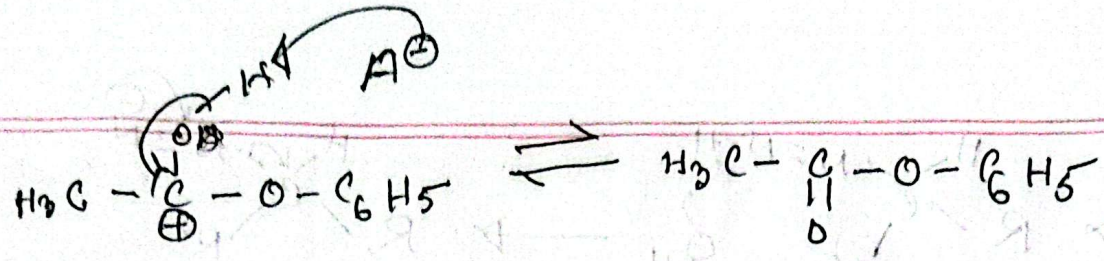
1. Yield কম → Yield বেশি-
2. দুইটায় Alkyl group থাকলে এ Process use করা যায় → Alkyl ও Aryl সবার ক্ষেত্রেই ব্যবহার্য।

Mechanism:

1) $R-\overset{O}{\overset{\|}{C}}-R' + NH_2NH_2 \rightleftharpoons R-\overset{N}{\overset{\|}{C}}-R'$ (with NH_2, H, OH) $\xrightarrow{-H_2O}$

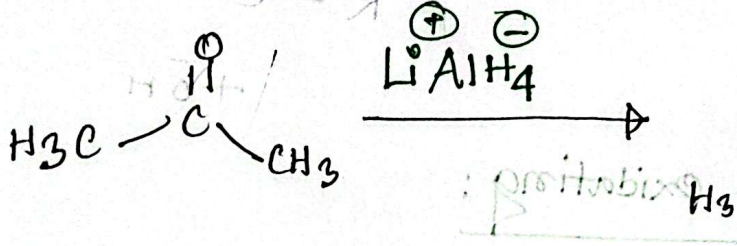
$$R-\overset{R'}{\underset{|}{C}}=N-\overset{-}{N}-H \longleftrightarrow R-\overset{\ominus}{C}(R)-N=N-H$$



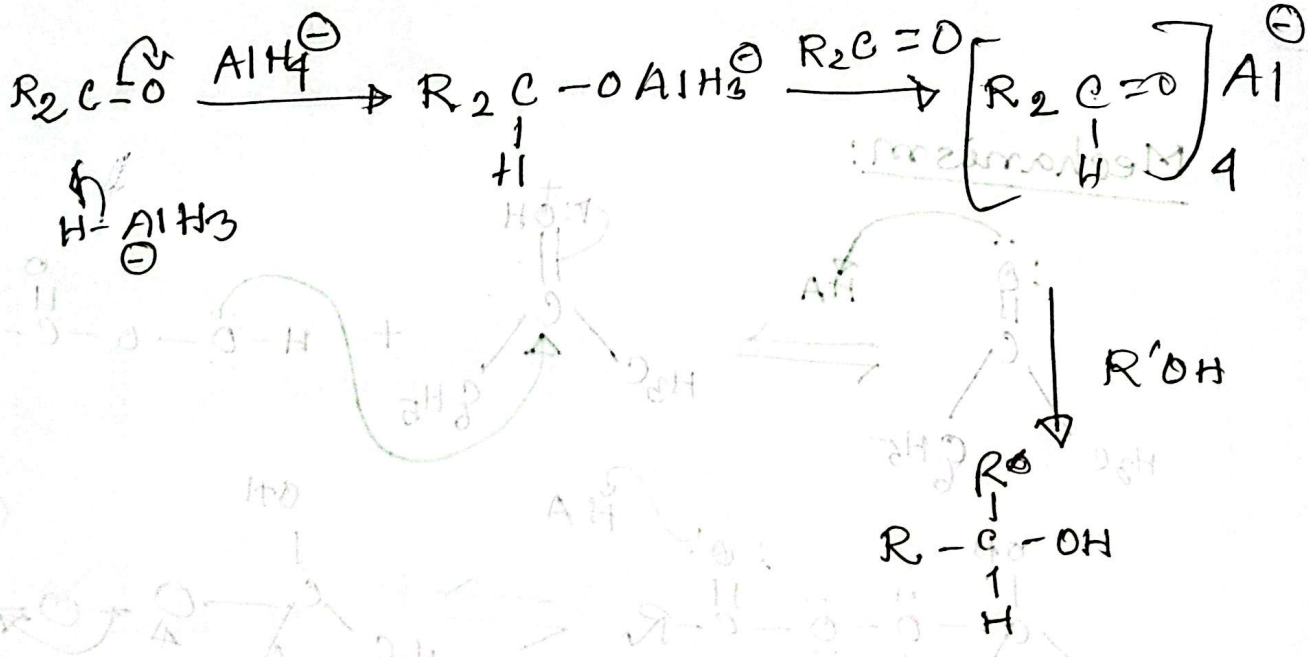


"O" কোথা থেকে আসছে তার প্রমাণ করার জন্য ^{18}O use করার জন্য যেতে পারেন।

Reduction by $LiAlH_4$ / $NaBH_4$:

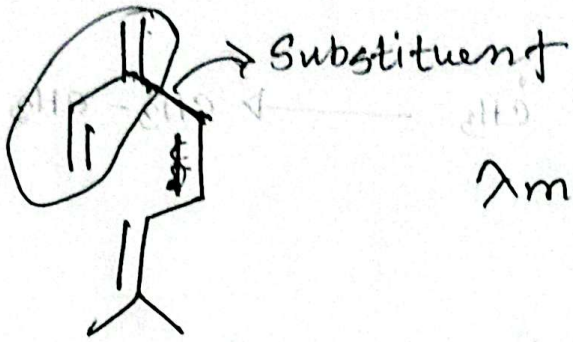


Mechanism:



*IR, NMR valomoto পড়েছো।
* Mass, UV – fundamental পড়িয়ে দিবো ভাই।
Auxochrome, chromophore, maglaparity. Mol. ion,

Woodward-Fieiser Rule:



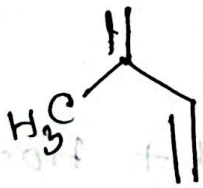
λmax

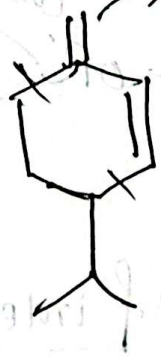
Myrcene

Base value = 214 nm

Substituent 2 টা nm ⟶ Alkyl substituent থাকলে
219 nm — তখন value টা fixed

λmax of S = 224 nm





214 nm

2×5 = 10 nm.

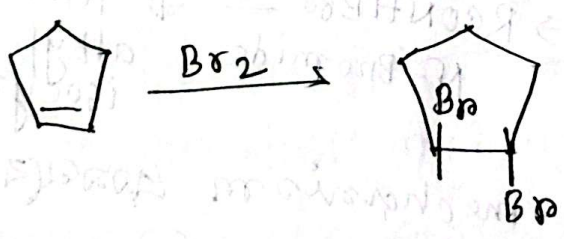
Stereochemical Criteria: F-22

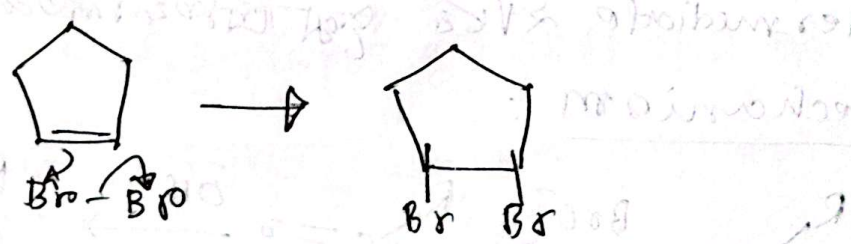
The base-catalysed bromination of an optically active stereoisomer of the ketone leads to an optically inactive

$$PhCO\overset{*}{C}HMeEt \xrightarrow[OH^-]{Br_2} PhCO\overset{*}{C}BrMeEt$$
$$(\pm)$$

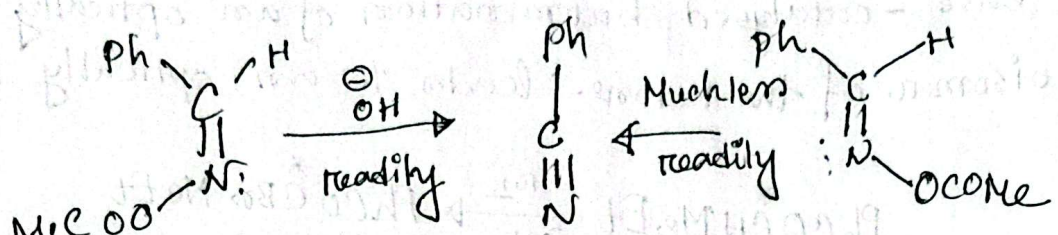
racemic product, indicates that the reaction must proceed through a planare intermediate, which can undergo attack equally well from either side leading to equal amounts of the two mirror-images form of the product.

Then, addition of Br_2 to cyclopentene, gives trans bromide, indicates that the mechanism of the reaction cannot be simply be direct. If it was one step, it should give cis-product. The addition must be at least two-step process.





Elimination:



Anti

syn.

Both Anti & syn give same cyanide as product, since it clearly sets limitation to which any mechanism advanced for the reaction will have to conform.

The study of intermediates:

(i) Separation of intermediates;

প্রক্রিয়া ক্ষেত্রে বা কখনো কোনো এক অবস্থায় intermediates product কে সুযোগ করা যায়। যেমন:

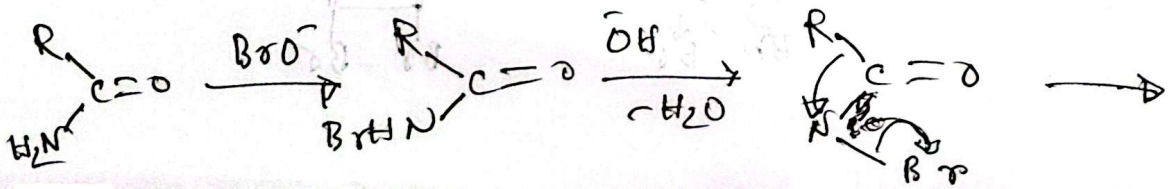
$$RCONH_2 \xrightarrow[Br_2]{KOH} RCONHBr \xrightarrow{KOH} RNCO \xrightarrow{H_2O} RNH_2 + CO_2$$

N-Bromide

alkyl isocyanate

এই প্রক্রিয়ার mechanism এভাবেই হতে পারে এবং এখানে intermediate গুলো পাওয়া সম্ভব।

Mechanism:

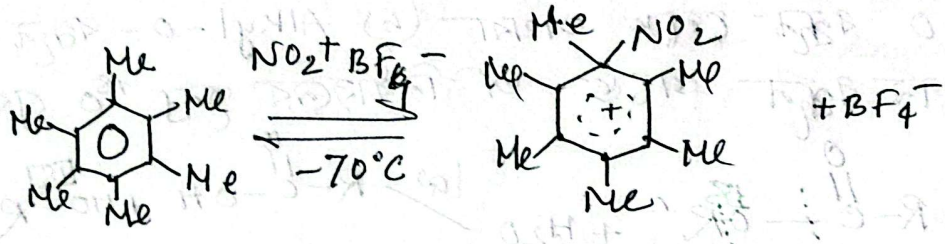


$$R-N=C=O \xrightarrow{H_2O} R-NH_2 + CO_2$$

(2) identification of intermediate product:

অধিকাংশ ক্ষেত্রে intermediate separate করা যায় না, তাই- IR, NMR, Mass spectroscopy use করা হয়।

NMR is used for (I)



(3) Trapping: F-22

যখন এ সমস্ত কোনোটাই করা যায়নি তখন trapping করা হয়। এসব ক্ষেত্রে প্রত্যাশিত intermediate যে যৌগসমূহ বিক্রিয়া করতে পারে বলে ধারণা করা হয় সেগুলি যোগ করা, তাদের পর্যবেক্ষণ করা। যেমন: Claisen rearrangement এ para migration হয়। এতে intermediate dienone তৈরি হয়। এটা ধরা যে, 1,3-Diene এর সাথে maleic anhydride সরাসরি যোগফলে যুক্ত যৌগ গঠন করে। সুতরাং, maleic anhydride সহযোগে claisen পুনর্বিন্যাস সম্পন্ন করলে এবং দেখা যায় যে প্রত্যাশিত যুক্ত যৌগ গঠন করে। সুতরাং এরা প্রমাণিত হয় যে, এখানে dienone তৈরি হয়েছিল। এই পদ্ধতিকে 'ফাঁদে ধরা' or Trapping বলে।

Isotope Using :

(i) Isotope lebelling : এই প্রক্রিয়ায় আইসোটোপ প্রবিষ্টকরণ করা হয় এবং উৎপাদে তার Presence পরিক্ষা করা হয়। উপস্থিতি/অনুপস্থিতির depend করে mechanism দেওয়া হয়।

$$RCOOR' + H_3O^+ \xrightarrow{H^+} RCOOH + R'OH$$

(a) Acyl-O বন্ধন ভেঙে অথবা (b) Alkyl-O- বন্ধন ভাঙে। কোন বন্ধন ভাঙে তা নির্ধারনের জন্য ^{18}O ব্যবহার করা হয়।

$$R-\overset{O}{\overset{\|}{C}}-\ddot{O}R' + H_2O \begin{cases} \xrightarrow{(a)} R-\overset{O}{\overset{\|}{C}}-OH + HO-R' \\ \xrightarrow{(b)} R-\overset{O}{\overset{\|}{C}}-OH + R'OH \end{cases}$$

বিক্রিয়ার পণ্য উৎপাদ বিশ্লেষন করলে দেখা যায় যে, ^{18}O যৌগে থাকে A পাওয়া যায় তাই acyl-O-বন্ধন ভাঙে এবং সেইভাবে mechanism নির্ধারন করা হয়।

(ii) Isotopic effect: কোনো পরমাণুর বন্ধন বিক্রিয়ায় participate করছে কিনা তা জানার জন্য তার আইসোটোপ ব্যবহার করে আমরা কাঠ না করে সেই যৌগের rate of reaction compare করি। যদি rate পরিবর্তন হয় তাহলে participate করছে এবং mechanism ও সেইভাবেই আগাই।

$$Ph_2C(OH)H \xrightarrow[\bar{O}H]{MnO_4^-} Ph_2C=O$$

C–H ~~strong~~ weaker

C–D stronger

$\frac{K_H}{K_D} = 7$ at 25°C

$\frac{K_H}{K_T} = 15$

* Q-2022 *

SUZON-50

(a) Repeat Question.

(b) Can you differentiate the possible isomers of a compound having molecular formula C_2H_6O with the help of IR spectroscopic data? If possible explain it.

Ans:

Molecular Formula C_2H_6O

The isomers of it,

CH_3-O-CH_3 CH_3CH_2OH

Dimethyl ether ethanol.

Differentiation betn the isomers using IR:

$\bar{\nu}$ max (cm^{-1})	Inference
$\bar{\nu}_{C-H\ (sp^3)} = 2800-2900\ cm^{-1}$	stretching vibration
$\bar{\nu}_{O-H} = 3500-3800\ cm^{-1}$ (Free O-H)	symmetric stretching.
$\bar{\nu}_{C-O} = 1260-1800\ cm^{-1}$	stretching vibration.
$\bar{\nu}_{C-O-H} = 1550-1220\ cm^{-1}$	C-O-H bending.
$\bar{\nu}_{O-H\ (H-bonded)} = 3400-3300$	stretching.

For Dimethyl ether.

$\nu_{max}(cm^{-1})$	Inference.
2960-2850 cm^{-1}	C-H (sp^3) Stretching vibration.
C-O = 1260-1000 cm^{-1}	(C-O) stretching vibration.

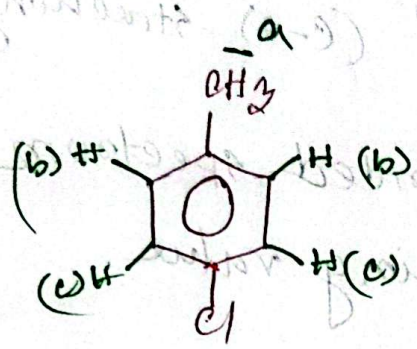
No hydrogen bonded spectrum / value

No O-H stretching value.

From the above information, we can see that. For ethanol we get many of the values of IR but for dimethyether we get C-H & C-O stretching only. Here, O-H = 3500-3650 cm^{-1} is missing. Thus, we can differentiate in this way. the above is one by IR.

(c) Draw the ^1H-NMR spectrum of p-chlorotoluene. Mention the approximate chemical shift value of each type of protons and explain the splitting pattern indicated in your spectrum.

Ans:



(a) $n+1 = 0+1 = 1$ singlet

so, δ 2.4 (3H, s)

(b) $n+1 = 1+1 = 2$ (doublet)

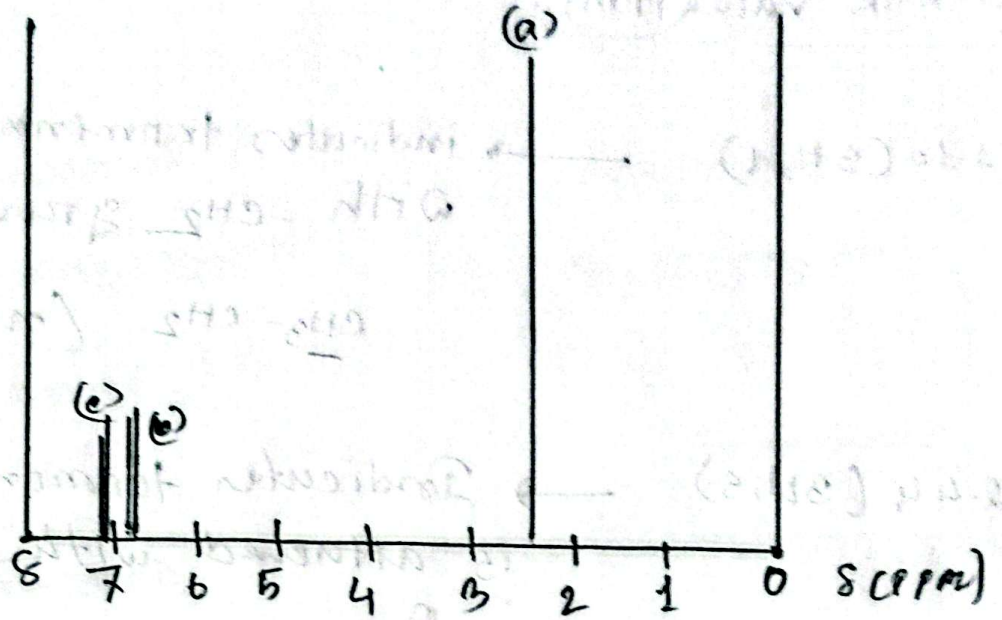
$\delta = \frac{7.10}{6.90}$ (2H, d)

(c) $n+1 = 1+1 = 2$ (doublet)

$\delta = \frac{6.90}{7.10}$ (2H, d)

Here, electron withdrawing -Cl atom is present so, deshield occurs and hence, more chemical shift we observe.

Now the spectrum for the compound :



(d) An organic compound of molecular formula $C_4H_8O_2$ shows the following spectral data :

IR : 2960, 2850, 1735, 1125 cm^{-1}

^1HNMR : δ 1.1 (3H, t), 1.44 (3H, s), 3.88 (2H, q)

Ans :

$\bar{\nu}_{(max)}$ IR :	Inference : (অনুমান/সিদ্ধান্ত)
2960	Asymmetric -C-H (sp^3) stretching vib
2850	symmetric -C-H (sp^3) " "
1735	>C=O stretching vibration for ester group —C(=O)—O

(b) Distinguish between propanal and propyl alcohol with the help of IR spectroscopic data.

Solⁿ:

Propanal

$CH_3CH_2\overset{O}{\overset{\|}{C}}-H$

Propyl alcohol

$CH_3CH_2CH_2OH$

For propanal,

$\bar{\nu}_{max}$ value:	Inference
2800 - 2900	C-H (sp^3) streching vibration.
2700 - 2800	C-H of aldehydic streching vibrati
1465 cm^{-1}	CH_2 - bending
1375	CH_3 bending
1725 - 1740 cm^{-1}	C=O streching vibration.

For propyl alcohol,

IR value	Inference,
2800 - 2900 cm^{-1}	Asymmetric C-H (sp^3) streching vib of CH_3.
1465 cm^{-1}	$-CH_2-$ bending
1375 cm^{-1}	$-CH_3$ bending.
3500 - 3600 cm^{-1} (Free H-bonded) 3200 - 3400 (H-bond)	-OH streching vibration,

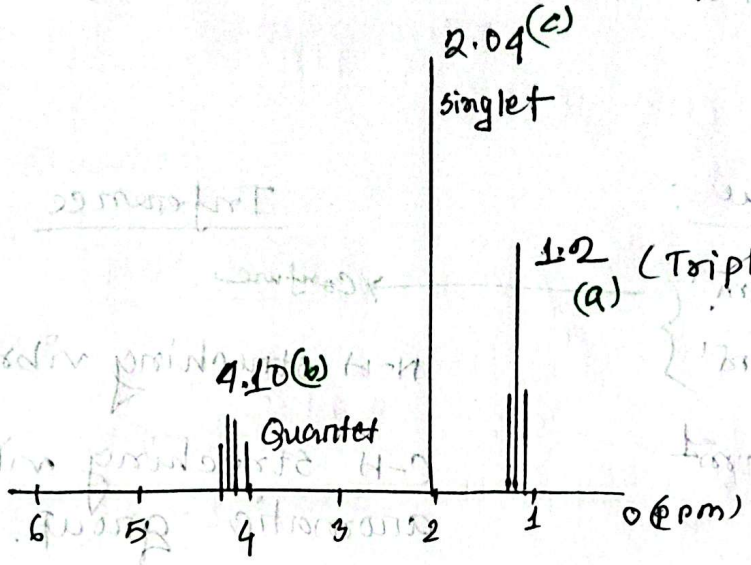
1000 – 1350 cm^{-1} C–O streching vibration.

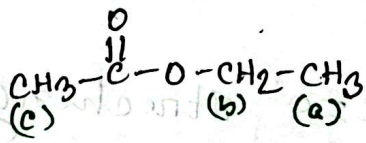
Aldhyde	vs	Alcohol
>C=O streching vibration given		No >C=O value. But it it shows C–O value.
No –OH streching value		It has O–H streching value 3500 – 3600 cm^{-1}.
It has aldehydic C–H value		But It has not.

So, by the above method, we can distinguish Proparnal & propylalcohol using IR.

(c) Draw the 1H-NMR spectrum of ethylacetate. Mention the approximate chemical shift value of each type of protons and explain the splitting patteron indicated in your spectrum.

Soln:





(a) δ_H (3H, triplet) → Indicates that, CH_3 is adjacent with $-CH_2$ group

(c) δ 2.04 (3H, s) → Indicates, CH_3 is adjacent with $-\overset{O}{\overset{\|}{C}}-O-$ group.

(b) δ 4.10 (2H, q) → Indicates $-CH_2-$ group is attached with $-CH_3$ group and $-O-\overset{O}{\overset{\|}{C}}-$ group.

(d) An organic compound of molecular formula C_7H_7ON shows the following data:

IR: 3400, 3350, 3080, 2960, 2850, 1680, 1600 cm^{-1}

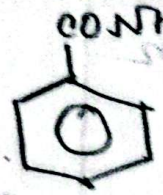
^1H-NMR: δ 7.2 – 7.3 (5H, m), 3.2 (2H, s)

Soln:

IR value:	Inference.
3400 cm^{-1}, 3350 cm^{-1}	→ confirms N-H streching vibrations.
3080 cm^{-1}	C-H streching vibration for aromatic group.
2960, 2850	
1680 cm^{-1}	>C=O streching which is adjacent with $-NH_2$ group
1600 cm^{-1}	C=C for benzene ring.

^1H-NMR	Inference.
δ 7.2 – 7.3 (5H, m)	which indicates these protons are in a benzene ring. It is multiplate.
δ 3.2 (2H, s)	Indicates, it is singlet and NH_2 is adjacent with $-\overset{O}{\overset{\|}{C}}-$

So, the probable structure :



$\equiv C_7H_7ON$

Benzamide

(a) Repeat Question

NMR Value

*Alkanes :

$R-CH_3$ 0.7 – 1.3 ppm

$R-CH_2-R$ 1.2 – 1.4 ppm

R_3CH 1.4 – 1.8 ppm

*Alkenes :

$C=C-H$ 4.5 – 6.5 ppm

$C=C-C-H$ 1.6 – 2.6 ppm

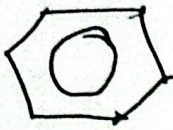
*Alkynes

$C\equiv C-H$ 1.7 – 2.7 ppm

$C\equiv C-CH$ 1.6 – 2.6

Aromatic compounds:

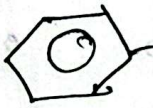
 6.5 – 8.0 ppm

 2.3 – 2.7 ppm
or
or $Ar-CH_3$ 2.4 – 2.7

Alcohols

C – OH 2.5 – 5.0 ppm

CH – O – H 3.2 – 3.8 ppm
(Ether/alcohol)

 OH δ = 8.30

Carboxylic Acids:

R – COOH 11 – 12 ppm

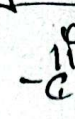
CH – COOH 2.1 – 2.5 ppm

Aldehydes

R – CHO 9.0 – 10.0 ppm

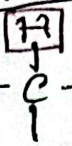
R – CH – CHO 2.1 – 2.4 ppm

Methyl ketone:

 2.0 – 2.4

 $Ar-CH_3$ 2.4 – 2.7

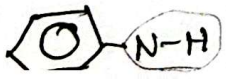
Alkyl halide

 2.5 – 4.0

Amine

R – N – H 0.5 – 4.0 ppm

– CH – N 2.2 – 2.9 ppm

 3.0 – 5.0 ppm

Red shift: A shift of absorption to a longer wavelength is called a bathochromic shift or red shift.

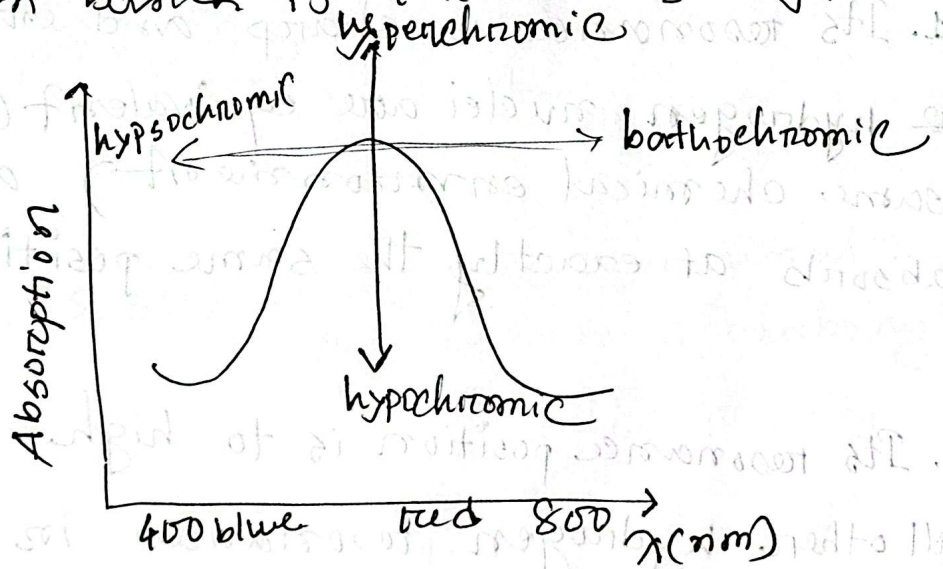
($\pi \longrightarrow \pi^*$)

Blue shift: A shift of absorption to a shorter wavelength is called hypsochromic shift or blue shift.

hypso-

Hyperchromism: Increased conjugation usually results in increased intensity is termed as hyperchromism.

Hypochromism: A decrease in intensity of an absorption band is termed as hypochromism.

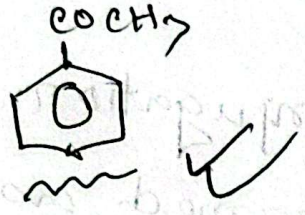


(*) 201b

(e) M.F C_8H_8O

IR: 3080, 2960, 2850, 1690, 1650 cm^{-1}

^1H-NMR: δ 7.2 – 7.3 (5H, m), 2.2 (3H, s)



TMS is considered as standard:

Reasons:

1. Its resonance is sharp and intense since all 12 hydrogen nuclei are equivalent (i.e have the same chemical environment) and hence absorb at exactly the same position.

2. Its resonance position is to high field of almost all other hydrogen resonances in organic molecules and hence easily can be recognized

3. It is a low b.p. liquid so can be readily removed from most samples after use.

2015: $C_6H_8O_3$

(i) UV: 283 nm ε_{max} 22

(ii) ~~$C_6H_8O_3$~~

IR value (cm^{-1})

2500-3000 (br) ⇒ ν_{OH} (carboxylic acid, str. symmetric)

1715 (s) cm^{-1} ⇒ $C=O$ streching vibration.

1342 cm^{-1} (w) ⇒ CH_3 bending

(iii)

δ 2.12 (S, 3H) → Indicates terminal $-CH_3$ adjacent to a carbonyl group.

$CH_3-C(=O)-$

δ 2.25 (t, 2H) → Indicates CH_2 group adjacent to a carbon that has two Hydrogen.

$HO-C(=O)-CH_2-CH_2-$

δ 2.60 (t, 2H) ⟶ Indicates $-CH_2-$ group adjacent to a carboxylic group and a methylene group

$$-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-OH$$

δ 11.11 (1H, s) ⟶ Indicates $-\overset{O}{\overset{\|}{C}}-OH$ group

So, the probable structure is,

$$CH_3-\overset{O}{\overset{\|}{C}}-\underset{2.60}{CH_2}-\underset{2.25}{CH_2}-\overset{O}{\overset{\|}{C}}-OH$$

* Woodward Fiesen rule *

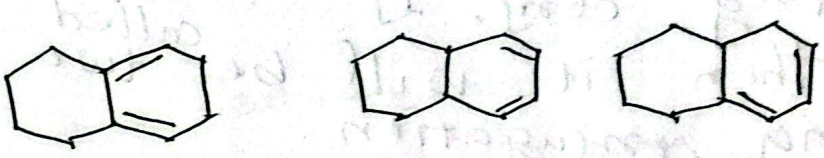


cis
cisoid

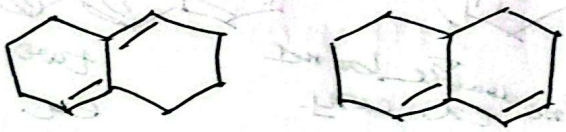


trans (taansoid)

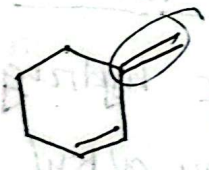
Homoannular diene:

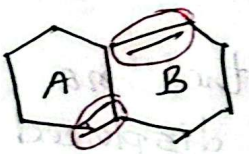


Heteroannular diene: Two double bonds are in two rings.

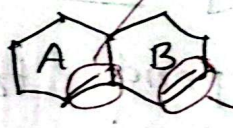


Exocyclic double bond; A double bond out side a ring.





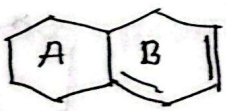
1



exocyclic to B

this double bond is not exocyclic to A

2



3



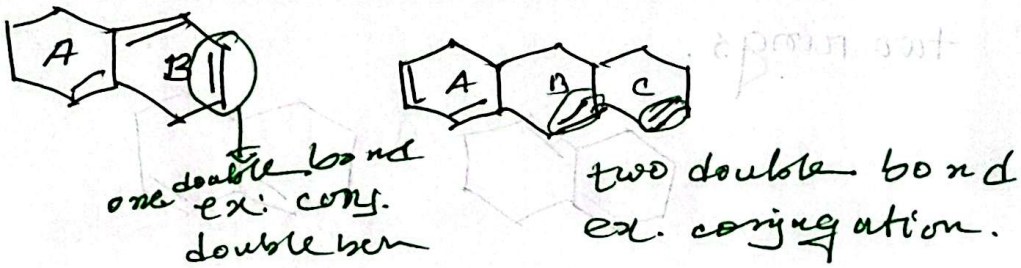
Both exocylic to A

4

⊞ Double bond extending conjugation:

In a conjugated diene, there are two double bonds, if it has one more double bonds in conjugation with other two then the additional double bond is called double bond extending conj. If there are two such bonds then it will be called two double bonds extending conjugation.

Ex:

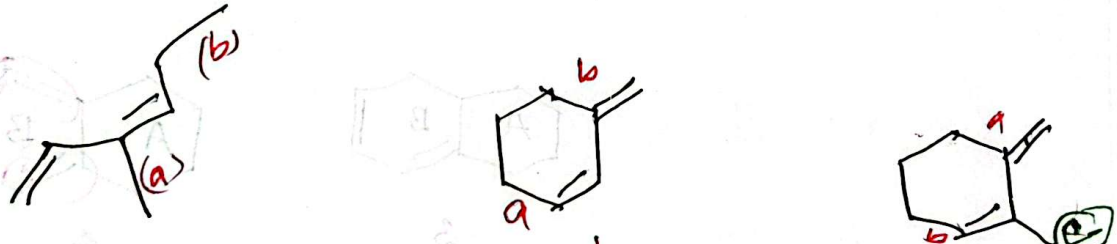


one double bond ex. conj. double bond

two double bond ex. conjugation.

⊞ Alkyl substituent or ring residue:

Corresponds to the no. of hydrogen atoms of conjugated system displaced by alkyl group or ring residue.

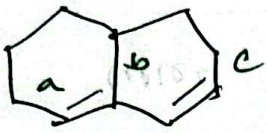


(A) 2H displaced by methyl & ethyl

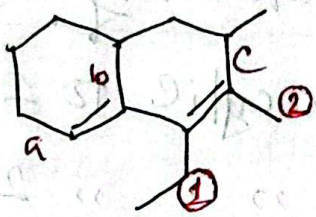
(B) 2H ....

(C)

(c) 2H are displaced by ring residue and another one is replaced by Methyl (c)

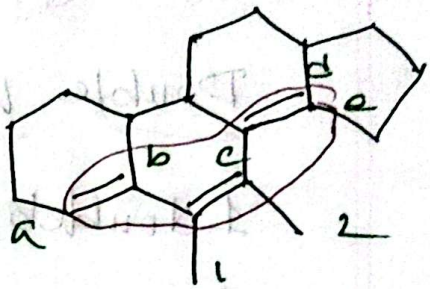


D



E

3H → ring
2H → ethyl, methyl groups



F

5H → ring
2H → Methyl groups

Woodward - Fieser's Rules for conj. dienes:

Base value

Acyclic conj. diene → 217 nm
Heteroannul " " → 214 nm
Homo " " " → 253 nm

Addition for each substituent

-R (Alkyl on ring residue): 5 nm
-OR (Alkoxy) → 6 nm
-SR (Thioether) → 30 nm
-X (Cl and Br) → 5 nm

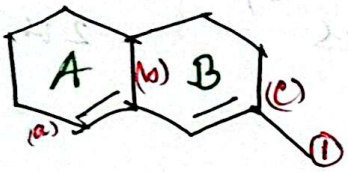
-OCOR (Acyloxy) : 0 nm

$-NR_2$ (Secondary amine) : 60 nm

Double bond extending conj. : 30 nm

1 doublebond exocylic to 1 ring : 5 nm

1 " " 2 " : 10 nm



Heteroannular diene, so base value (214) nm

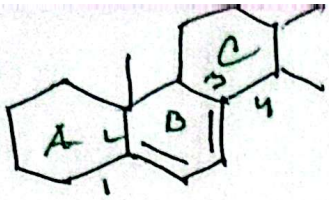
= 214 nm

Add for 3 ring residue = 3 × 5 "

" " 1 alkyl group = 5 nm

Add 1 exocylic double bond = 5 nm

λ_{max} = 239 nm



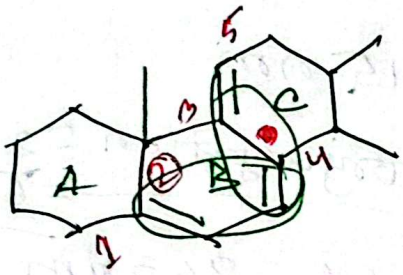
Homoannular diene, so base value = 253 nm

Add 4 ring residue = 4×5 nm

Add for 2 exocyclic double bond = 2×5=10 nm

λmax = 253+20+10

= 283 nm



Homo & Hetero annular diens both are present.

In this situation Homo annular base value should be considered.

Base value = 253

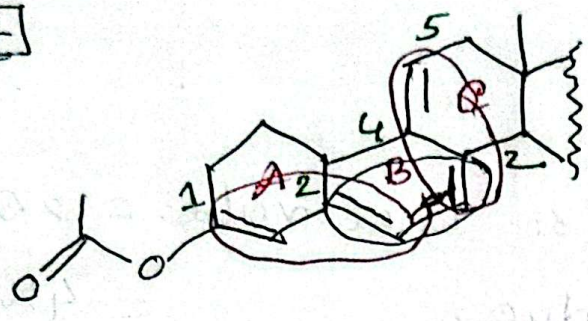
Add for 5 ring resid = 5×5=25nm

Add for 3 exocyclic double bond = 3×5=15nm

Add for extended conj. = 30nm

λmax = 323 nm.

*



Base value = 253

Add 5 ring residue = 5×5 = 25

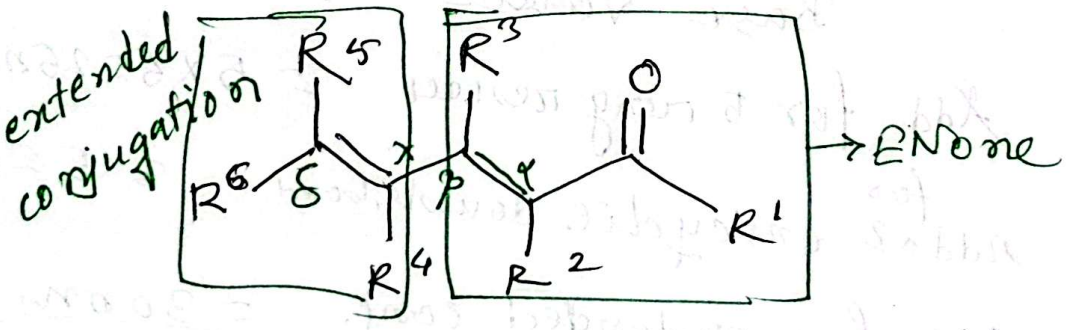
Add for 1 acetoxy group = 0

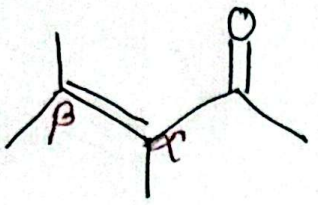
Add for 3 exocyclic double bond = 3×5 = 15 nm

Add for 2 extended conjugation = 2×30 = 60 nm

λ_{max} = 253 + 25 + 15 + 60 = 353 nm

Woodwood Fiesr rules for Enones:



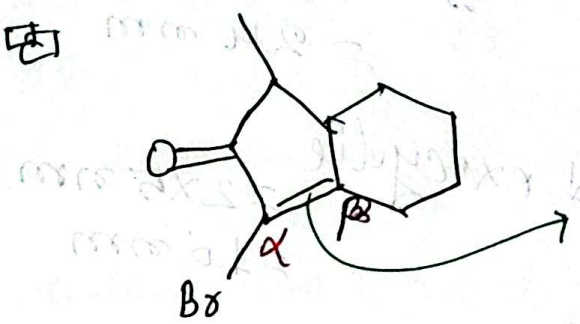


Acyclic α, β - unsaturated ketone : 215 nm

1α - Methyl : 10 nm

2β - methyl : 2×12 = 24 nm

λ_{max} = 249 nm



5 - membered ring cyclic α, β - unsaturated ketone :

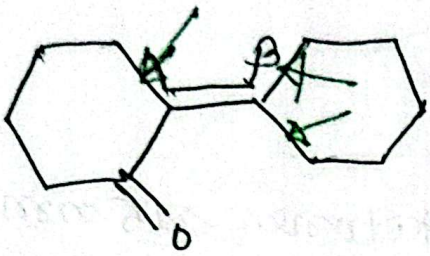
so base value = 202 nm

1α - Br = 25 nm

2β - ring residues = 2×12 nm

1 - Exocyclic double bond = 5 nm

λ_{max} = 256 nm.



6-member cyclic α,β-unsaturated ketone

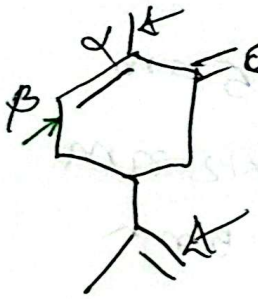
= 215 nm

1α-Ring residue = 10 nm

2β- " " = 2×12 nm
= 24 nm

1 double bond exocyclic = 2×5 nm
two rings = 10 nm

λ_{max} = 259 nm

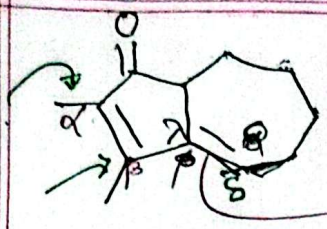


is not present as conjugation.

6-mem — = 215 nm

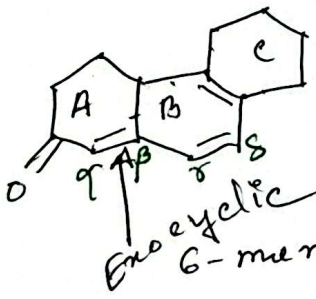
1α-alkyl = 10 nm

1β ring .... = 12 nm



→ Exocyclic

5-membered ring :	202 nm
1α- methyl	10 nm
1β- methyl	12
~~1β- ring residue~~	~~12~~
1γ- ring residue	18 nm
1δ- " "	18 nm
1- Exocyclic double bond	5 nm
Double bond extending conjugation	+30 nm
	λ_{max} = 295 nm.



Exocyclic 6-membered,	base value = 215 nm
β- substituent	= 12 nm
Higher position	= 3×18 nm
1 Homodiene	= 39 nm
2 DEC	= 30×2 = 60 nm
1 Exocyclic double bond	= 5 nm
	385 nm